

SRAI EDUCATIONAL WORKBOOK

Mathematical Thinking and Reproducible Computation

Exercises, capstone, worked solutions and assessment rubric

CENTRAL QUESTION Can another person reconstruct the analysis, verify what was calculated and judge whether the model is adequate for the intended decision?

Control	Value
Production unit	PU-B01-C01
Book alignment	SRAI Book 1 — Mathematical Foundations, Chapter 1
Notebook	M1_N01_mathematical_thinking.ipynb
Audience	Advanced learners, analysts, statisticians and analytical managers
Estimated effort	4–6 hours, excluding optional extension
Version / status	v1.0 — Draft for owner review

How to use this workbook

Complete Part I before consulting the worked solutions in Part II. The exercises move from conceptual identification to implementation, empirical fitting and decision consequences. Use the canonical notebook M1_N01 as the starting computational record, and save extensions under a new filename.

Learning outcomes

- Distinguish reality, abstraction, model, algorithm, implementation, verification, interpretation and decision.
- State variables, parameters, assumptions, objectives, constraints and omissions.
- Create a reproducible computational record with a controlled environment and random seed.
- Verify numerical expectations and test model sensitivity.
- Separate computational verification from empirical and operational validation.
- Analyse false alarms, missed events and accountability in a threshold decision.

Required submission

- Completed written responses for Exercises 1–4.
- One executed notebook containing Exercises 5–8 and all outputs.
- A one-page decision note for the capstone.
- A reproducibility record: Python version, package versions, seed, execution date and filenames.

Assessment map

Part	Focus	Points
A	Concepts and model anatomy	20
B	Reproducible implementation	25
C	Model extension and fitting	25
D	Decision-alert capstone	30

Integrity rule Do not alter or remove a failing assertion merely to obtain a clean run. Explain the failure, correct the cause and preserve evidence of the change.

Part A — Concepts and model anatomy

Exercise 1 — Model or reality?

The notebook represents electricity demand as $D(T) = \beta_0 + \beta_1 T$, with $\beta_0 = 120$ and $\beta_1 = 2.5$.

1. Identify the input variable, output variable and two parameters.
2. State the principal linearity assumption in ordinary language.
3. Name four relevant features of real electricity demand that the model omits.
4. Explain why an omitted feature is not automatically an error, but must be judged against the intended use.

Exercise 2 — Reconstruct the analytical chain

For an institution considering a heat-related capacity alert, complete the following chain.

Stage	Your response
Reality	
Abstraction	
Model	
Algorithm	
Python implementation	
Verification	
Interpretation	
Decision	

Exercise 3 — Verification or validation?

Classify each item as primarily verification, validation, or decision/governance evidence. Briefly justify each classification.

Evidence item	Classification and justification
The predicted values equal [170.0, 182.5, 195.0].	
A one-degree increase changes modeled demand by 2.5 units.	
Errors are larger during humid weeks.	
The model performs poorly in remote service zones.	
The accountable director approves a restricted pilot.	

Exercise 4 — Challenge an AI-generated answer

An AI assistant states: “The verified model proves that demand will rise by 2.5 units for every additional degree.” Write a professional response identifying at least three problems with this claim and the evidence required before operational use.

Part B — Reproducible implementation

Exercise 5 — Establish the computational record

5. Create a copy of M1_N01 and rename it `PU-B01-C01\_exercises\_<your\_name>.ipynb`.
6. Run the environment-information and seed cell before any analytical cell.
7. Record Python, NumPy and notebook-kernel versions.
8. Restart the kernel, run all cells and confirm that the outputs are unchanged.
9. Write one paragraph explaining what the seed controls and what it does not guarantee.

Expected evidence Executed environment output, seed 42, an uninterrupted Run All, and a written reproducibility note.

Exercise 6 — Add boundary and invariant tests

Add tests for the following behaviours. Explain why each test is relevant.

- At $T = 0$, predicted demand equals the baseline.
- Increasing T by 1 changes predicted demand by β_1 .
- A vector input returns a vector of the same length.
- A repeated call with the same input returns the same output.
- Decide how the model should handle a missing or non-numeric temperature, then test that behaviour.

```
assert np.allclose(model.predict([0.0]), [120.0])
assert np.allclose(model.predict([31.0]) - model.predict([30.0]), [2.5])
# Add the remaining tests and a short explanation.
```

Exercise 7 — Sensitivity table

Calculate predictions for temperatures from 15°C through 40°C in 5-degree increments. Present the results in a labelled table. Then repeat the calculation for $\beta_1 = 1.5, 2.5$ and 3.5.

10. Which parameter changes the slope?
11. Which parameter changes the intercept?
12. At what temperature is the difference between the lowest and highest sensitivity scenarios largest?
13. What information would be needed to determine which sensitivity is empirically credible?

Part C — Extend and estimate the model

Exercise 8 — Add humidity

Extend the model to $D(T,H) = \beta_0 + \beta_1 T + \beta_2 H$, where H is relative humidity measured in percentage points. Use $\beta_2 = 0.4$ for the initial experiment.

14. Implement an immutable model class with explicit parameter names.
15. Predict demand for (T,H) pairs $(20,40)$, $(25,60)$ and $(30,80)$.
16. Verify each prediction with independently calculated expected values.
17. Test the change produced by one additional degree at fixed humidity.
18. Test the change produced by one additional humidity point at fixed temperature.
19. List at least five limitations that remain after adding humidity.

Exercise 9 — Fit synthetic observations

Generate 100 synthetic observations using seed 42, temperatures between 15°C and 40°C, humidities between 30% and 90%, and normally distributed noise. Fit a linear regression and compare estimated coefficients with the data-generating values.

Required output	Acceptance check
Data-generation statement	Seed, distributions, sample size and true coefficients recorded
Coefficient table	True and estimated values shown together
Residual summary	Mean, standard deviation and largest absolute residual
Diagnostic graph	Observed versus predicted values with labels
Interpretation	Sampling variation and model assumptions discussed

Caution Good recovery of coefficients from synthetic data validates the exercise design under its own assumptions. It does not validate the model for real electricity systems.

Part D — Decision-alert capstone

Scenario

A system operator considers issuing a capacity alert whenever predicted demand exceeds 190 units. An alert triggers costly standby generation. A missed high-demand event risks service interruption, with potentially unequal effects across locations and population groups.

Task 1 — Define the decision

20. State the decision, target population or service area, time horizon and accountable owner.
21. Define a “high-demand event” independently of the model prediction.
22. Identify the people or areas most affected by false alarms and missed events.
23. State at least three operational or legal constraints.

Task 2 — Evaluate threshold performance

Using a synthetic or instructor-provided evaluation dataset, calculate the confusion matrix for thresholds 180, 190 and 200.

Threshold	True alerts	False alerts	Missed events	Correct non-alerts
180				
190				
200				

24. Calculate precision, recall and false-alarm rate for each threshold.
25. Explain why no single metric is sufficient for the decision.
26. Assign explicit relative costs to a false alert and a missed event; justify them.
27. Recommend a threshold, restricted use, further testing or rejection.

Task 3 — Stress test

- Evaluate performance by temperature range, humidity range, geography or another meaningful subgroup.
- Test a plausible shift in the input distribution.
- Identify at least one condition under which alerts should be suspended.
- Describe the monitoring evidence required after deployment.

Task 4 — One-page decision note

Write for a named senior decision-maker. Include: recommendation; evidence; assumptions; reproducibility status; verification status; validation limits; false-alarm and missed-event consequences; affected groups; safeguards; monitoring; and accountable owner.

Capstone standard A polished prediction is not the final product. The final product is a controlled, inspectable and appropriately limited decision recommendation.

Part II — Worked solutions and instructor guidance

Instructor section Learners should complete Part I before consulting these solutions. Equivalent reasoning and independently correct code should receive credit.

Solutions 1-4 — Concepts

Exercise 1. T is the input; D is the output; β_0 is baseline demand and β_1 is temperature sensitivity. The model assumes a constant linear change over the represented range. Omissions include humidity, seasonality, day type, economic activity, prices, outages and nonlinear thresholds. An omission is acceptable only when it does not undermine the intended use or is explicitly treated as a limitation.

Exercise 2. A defensible chain begins with the service system and alert decision, abstracts relevant demand drivers, states variables and constraints, defines the predictive and threshold procedures, implements them reproducibly, verifies code behaviour, interprets uncertainty and consequences, and leaves authorization with a named owner.

Exercise 3. Expected numerical outputs and slope checks are verification evidence. Performance during humid weeks and in remote zones is validation evidence. Director approval is decision/governance evidence. None of these categories substitutes for the others.

Exercise 4. The AI claim confuses model behaviour with reality, generalizes beyond the tested range and uses “proves” without empirical validation. Operational use requires relevant observed data, external and temporal tests, subgroup analysis, uncertainty, consequence analysis and accountable authorization.

Solutions 5-7 — Reproducibility and sensitivity

Exercise 5. A complete record contains the environment, dependencies, seed, controlled inputs, execution order and outputs. Seed 42 controls compatible pseudo-random operations; it does not guarantee identical results across every library, hardware platform, parallel operation or external service.


```
temps = np.arange(15.0, 41.0, 5.0)
for slope in [1.5, 2.5, 3.5]:
    scenario = LinearDemandModel(120.0, slope)
    print(slope, scenario.predict(temps))
```

Exercise 7. β_1 changes slope and β_0 changes intercept. The gap between slope scenarios grows with temperature because the difference is proportional to T . Observed, quality-controlled data and an appropriate validation design are required to judge empirical credibility.

Solution 8 — Humidity extension

```
@dataclass(frozen=True)
class DemandWithHumidity:
    baseline: float
    temperature_sensitivity: float
    humidity_sensitivity: float

    def predict(self, temperature_c, humidity_pct):
        t = np.asarray(temperature_c, dtype=float)
        h = np.asarray(humidity_pct, dtype=float)
        return self.baseline + self.temperature_sensitivity*t + self.humidity_sensitivity*h

model_th = DemandWithHumidity(120.0, 2.5, 0.4)
result = model_th.predict([20,25,30], [40,60,80])
expected = np.array([186.0, 206.5, 227.0])
assert np.allclose(result, expected)
```

At fixed humidity, one additional degree changes modeled demand by 2.5 units. At fixed temperature, one additional humidity point changes it by 0.4 units. Remaining limitations include seasonality, time of day, prices, outages, economic activity, interactions and nonlinear responses.

Solution 9 — Synthetic fitting guidance

Estimated coefficients should be reasonably close to the generating values, but exact equality is neither expected nor required because the observations contain noise. Assessment should focus on transparent generation, correct fitting, comparison with true values, residual inspection and a restrained interpretation.

Capstone guidance

The preferred threshold cannot be determined from accuracy alone. It depends on event prevalence, the comparative costs of false alerts and missed events, operational capacity, subgroup effects and

tolerance for risk. Full credit requires an explicit decision rule, transparent cost assumptions, validation limits, safeguards, monitoring and named accountability. A justified recommendation to delay deployment may be stronger than an unsupported threshold choice.

Assessment rubric

Criterion	Excellent	Adequate	Needs revision	Points
Model reasoning	Variables, assumptions, objectives, constraints and omissions are precise.	Most elements identified; minor gaps.	Model confused with reality or key assumptions missing.	20
Reproducibility	Environment, seed, inputs, code and outputs reconstruct cleanly.	Run mostly reproducible; limited metadata gaps.	Hidden state, missing files or unexplained output.	20
Verification	Independent expectations, boundary tests and invariants are convincing.	Core assertions present.	Relies mainly on code running without error.	15
Validation reasoning	External adequacy, subgroups, shifts and limitations treated correctly.	Main distinction understood.	Verification treated as validation.	15
Decision analysis	Costs, errors, affected groups, safeguards and owner integrated.	Recommendation supported but incomplete.	Threshold chosen without consequence analysis.	20
Communication	Clear, concise and professionally documented.	Understandable with minor ambiguity.	Evidence and recommendation difficult to follow.	10

Source and control note

Developed from SRAI Book 1, Chapter 1, “Mathematical Thinking and Reproducible Computation,” and canonical notebook M1_N01. This standalone asset expands the four embedded notebook prompts into a complete learner workbook, capstone, solutions section and assessment rubric. It remains a controlled draft until Mbaye Kebe completes substantive review and approval.